

G3PB-III closure: the physical 200/110 amplitude ratio lies in the no-condensate-certified box

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

G3PB-III is the residual G3'-b(iii) registered by Math432: “extraction of the actual $\{200\}/\{110\}$ amplitude ratio from the Math400-AddF N=64 L-BFGS-B states (desirable cross-check; low priority since the entire scanned (A_1, A_2) plane is positive).” The premise has since been upgraded from a grid statement to a proven exact-Wick CONTINUUM no-condensate over the whole (A_1, A_2) box at the B1 point $r_{\text{bare}} = 0.219$ (twoshell-anchored-continuum v1.0). This note supplies the cross-check: it extracts the physical $\{200\}/\{110\}$ ratio and confirms the physical operating point is covered.

2. The physical $\{200\}$ response and the cross-check

Using the exact Math432 engine (neuter-imported, validated to 5×10^{-8} against Math432’s recorded brackets), the physical $\{200\}$ response is $A_2^*(A_1) = \arg \min_{A_2, M} \Delta F_{\text{anchored}}(A_1, A_2, M)$ – the amplitude the secondary $\{200\}$ shell takes when driven by the $\{110\}$ primary:

A_1	A_2^*	$\rho = A_2^*/A_1$	$\Delta F_{\text{anchored}}$ (per-row min)	in box
0.06	−0.016	−0.267	$+1.14 \times 10^{-2}$	yes
0.08	−0.016	−0.200	$+2.31 \times 10^{-2}$	yes
0.10	−0.048	−0.480	$+4.37 \times 10^{-2}$	yes
0.12	−0.064	−0.533	$+7.90 \times 10^{-2}$	yes
0.14	−0.080	−0.571	$+1.40 \times 10^{-1}$	yes

The response is on the NEGATIVE branch (sextic-driven, consistent with Math432’s tilt-sign observation and Math434’s audited per-row optimum $A_2 \approx -0.065 \pm 0.015$ at $A_1 = 0.14$), with $|A_2^*| \leq 0.08$ and $|\rho| \leq 0.57$: the $\{200\}$ shell is suppressed relative to $\{110\}$ and the physical trajectory lies well interior to the continuum-certified box $|A_1|, |A_2| \leq 0.16$. Along the physical-ratio trajectory $\Delta F_{\text{anchored}} > 0$ for every A_1 , so Reading-H wins even at the most-favourable secondary-shell response.

3. Closure

The physical $\{200\}/\{110\}$ ratio is extracted and confirmed to lie inside the no-condensate-certified box, where the exact-Wick anchored free energy is positive (continuum, twoshell-anchored-continuum v1.0). G3PB-III (G3'-b(iii)) is CLOSED at CROSS-CHECK grade (operator-directed 2026-06-08). With ESTIMATOR-UPGRADE (CLOSED@CONTROLLED-ERROR) and now G3PB-III closed, B1-RH-ENUM has NO open gates; it remains T6 CONDITIONAL on $\{\text{H-LAYER, H-ADM-COH, SC-SCOPE}\}$ (the gate governed an ensemble cross-check, not the selection SIGN nor a hypothesis).

4. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

- (α) “The ratio is from the engine, not from AddF N=64.” VALID-with-mitigation: the AddF raw L-BFGS-B states are not migrated to this repository, so the ratio is extracted from the exact $\{110\} + \{200\}$ engine – which IS the validated two-shell model whose $\{200\}$ response is the faithful in-truncation $\{200\}/\{110\}$ ratio. The cross-check is at the $\{110\} + \{200\}$ scope by construction; the full N=64 higher-shell content is the separately registered G3’-b(i)/(ii), not G3PB-III.
- (β) “Does a single probed A_1 grid miss the danger point?” DISMISSED: the cross-check does not rest on the probed rows alone – the whole (A_1, A_2) box is already a proven exact-Wick continuum no-condensate (twoshell-anchored-continuum v1.0, bulk curvature-chord $+1.3 \times 10^{-3} + \text{near-origin PD}$); the row table only LOCATES the physical ratio inside that box.
- (γ) “Does closing this change B1’s tier?” DISMISSED: G3PB-III is an ensemble-cross-check gate (does a richer condensate beat Reading-H?), answered NO; it is not one of B1’s named hypotheses $\{\text{H-LAYER, H-ADM-COH, SC-SCOPE}\}$. Closing it removes the last quality gate; the tier stays T6 CONDITIONAL.

Quantitative sanity check (mandatory): magnitude – $|A_2^*| \leq 0.08 \ll 0.16$ (box half-width), $|\rho| \leq 0.57$; sign-direction – $A_2^* < 0$ (negative branch, sextic-driven), $\Delta F_{\text{anchored}} > 0$ along the trajectory; limit-case – the $A_1 = 0.14$ optimum $A_2^* = -0.080$ sits near Math434’s audited -0.065 ± 0.015 ; consistency – every per-row min exceeds the diagonal whole-box min 3.9×10^{-5} , consistent with the continuum no-condensate.

5. Result footer

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Result ID:          B1-RH-ENUM / G3PB-III (G3'-b(iii)) CLOSED@CROSS-CHECK
Precise statement:  The physical {200} response A2*(A1)=argmin_{A2,M}
                    dF_anchored at r=0.219 is on the negative branch with
                    |A2*|<=0.08, |rho|<=0.57, inside the box |A1|,|A2|<=0.16;
                    dF_anchored>0 along the physical-ratio trajectory
                    (+1.1e-2..+1.4e-1). The {200}/{110} ratio does not open a
                    sub-Reading-H valley; combined with the whole-box exact-
                    Wick continuum no-condensate, Reading-H wins at the
                    physical ratio.
Scope:              {110}+{200} truncation (AddF N=64 raw not migrated; ratio
                    from the validated exact engine). Higher shells =
                    separate G3'-b(i)/(ii).
Dependencies:       Math432 engine (neuter-imported); twoshell-anchored-
                    continuum v1.0 (whole-box continuum no-condensate);
                    Math434 (audited per-row optimum cross-check).
Evidence grade:     EXECUTED (g3pb3_ratio_extraction.py 6/6).
Reproduction command: python codes/vacuum/g3pb3_ratio_extraction.py
Expected output:    6/6 PASS; physical ratio rows; |rho|<=0.57; in-box;
                    dF>0 along the trajectory.
Falsification gate: A physical A2*(A1) outside the box, or dF_anchored<0 along
                    the physical-ratio trajectory at r=0.219.
Tier before / after: G3PB-III: OPEN -> CLOSED@CROSS-CHECK. B1-RH-ENUM: T6
                    CONDITIONAL -> T6 CONDITIONAL (unchanged; ensemble cross-
                    check, not the SIGN). open_gates: [G3PB-III] -> [].
No-overclaim statement: CROSS-CHECK grade within the {110}+{200} truncation; the

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ratio is engine-extracted (AddF raw not migrated). The full N=64 higher-shell content is the separate G3'-b(i)/(ii). B1 T6 SIGN and its three hypotheses are unaffected. Next required action: None for G3PB-III (closed). B1 has no open gates; the remaining levers are the named hypotheses {H-LAYER, H-ADM-COH, SC-SCOPE}.